AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #4

Activity Name: LCD Interfacing

Name of Student:	**Chetan Kumar G**
Registration No.:	**CH.SC.U4CSE24109**
Activity Date:	**18|08|2026**
Submission Date:	**18|08|2026**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

Faculty In-charge

Write an embedded C program for ARM architecture to interface LCD. Debug and verify the program using IDE on a development computer system.

PROGRAM:

```
#include <LPC21xx.h>

#define LCD_DATA 0x000000FF      // P0.0 to P0.7
#define RS (1<<16)
#define EN (1<<17)

void delay(unsigned int t)
{
    unsigned int i,j;
    for(i=0;i<t;i++)
        for(j=0;j<600;j++);
}

void lcd_cmd(unsigned char cmd)
{
    IOCLR0 = LCD_DATA;
    IOSET0 = cmd;

    IOCLR0 = RS;        // Command mode

    IOSET0 = EN;
    delay(2);
    IOCLR0 = EN;

    delay(5);
}

void lcd_data(unsigned char data)
{
    IOCLR0 = LCD_DATA;
    IOSET0 = data;

    IOSET0 = RS;        // Data mode

    IOSET0 = EN;
    delay(2);
    IOCLR0 = EN;

    delay(5);
}
```

```
void lcd_string(char *str)
{
    while(*str)
    {
        lcd_data(*str++);
    }
}

void lcd_init()
{
    delay(20);

    lcd_cmd(0x38);   // 8-bit, 2-line
    lcd_cmd(0x0C);   // Display ON
    lcd_cmd(0x06);   // Cursor increment
    lcd_cmd(0x01);   // Clear display
}

int main()
{
    IODIR0 = LCD_DATA | RS | EN;

    lcd_init();

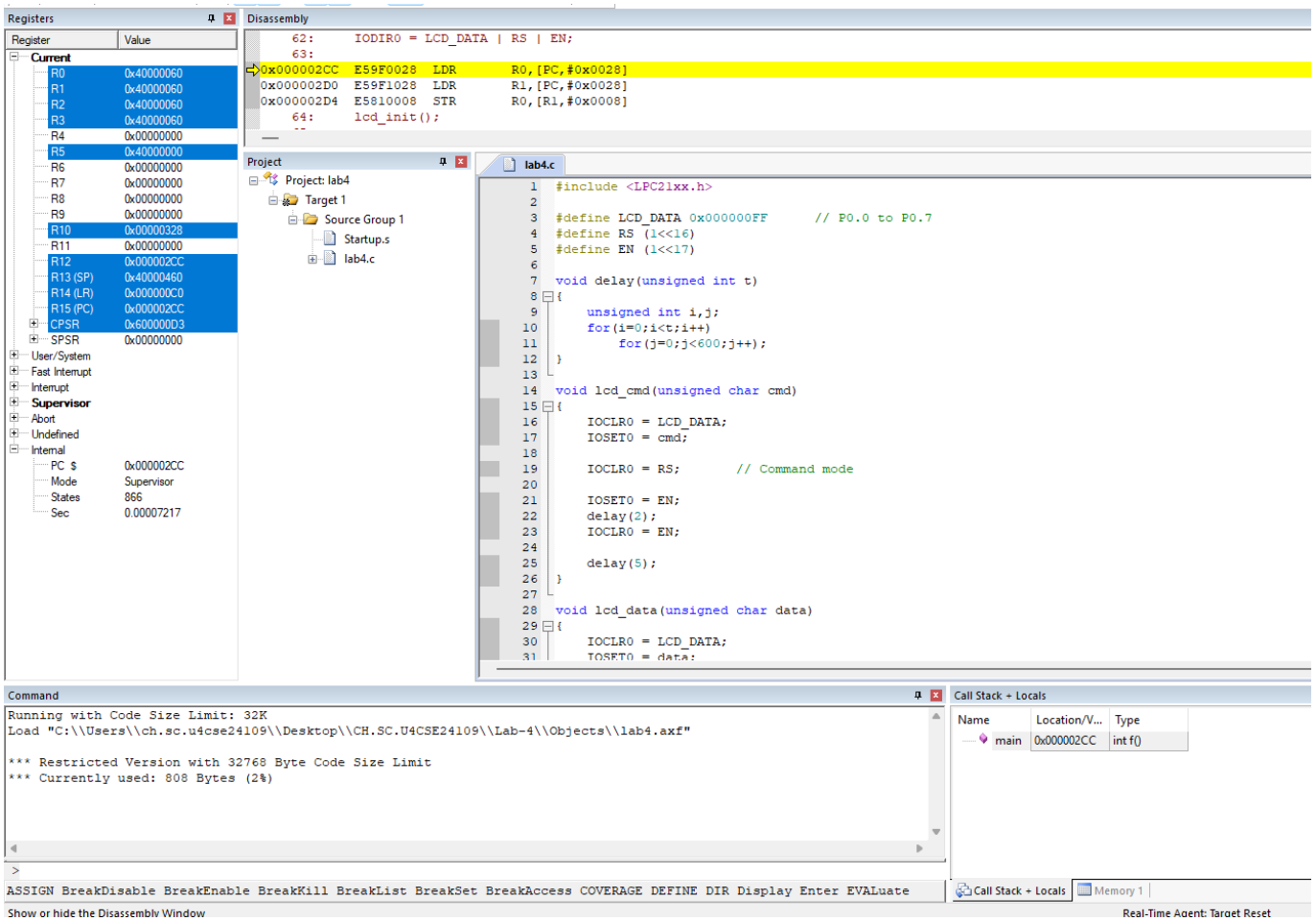
    lcd_string("HELLO");

    lcd_cmd(0xC0);      // Second line

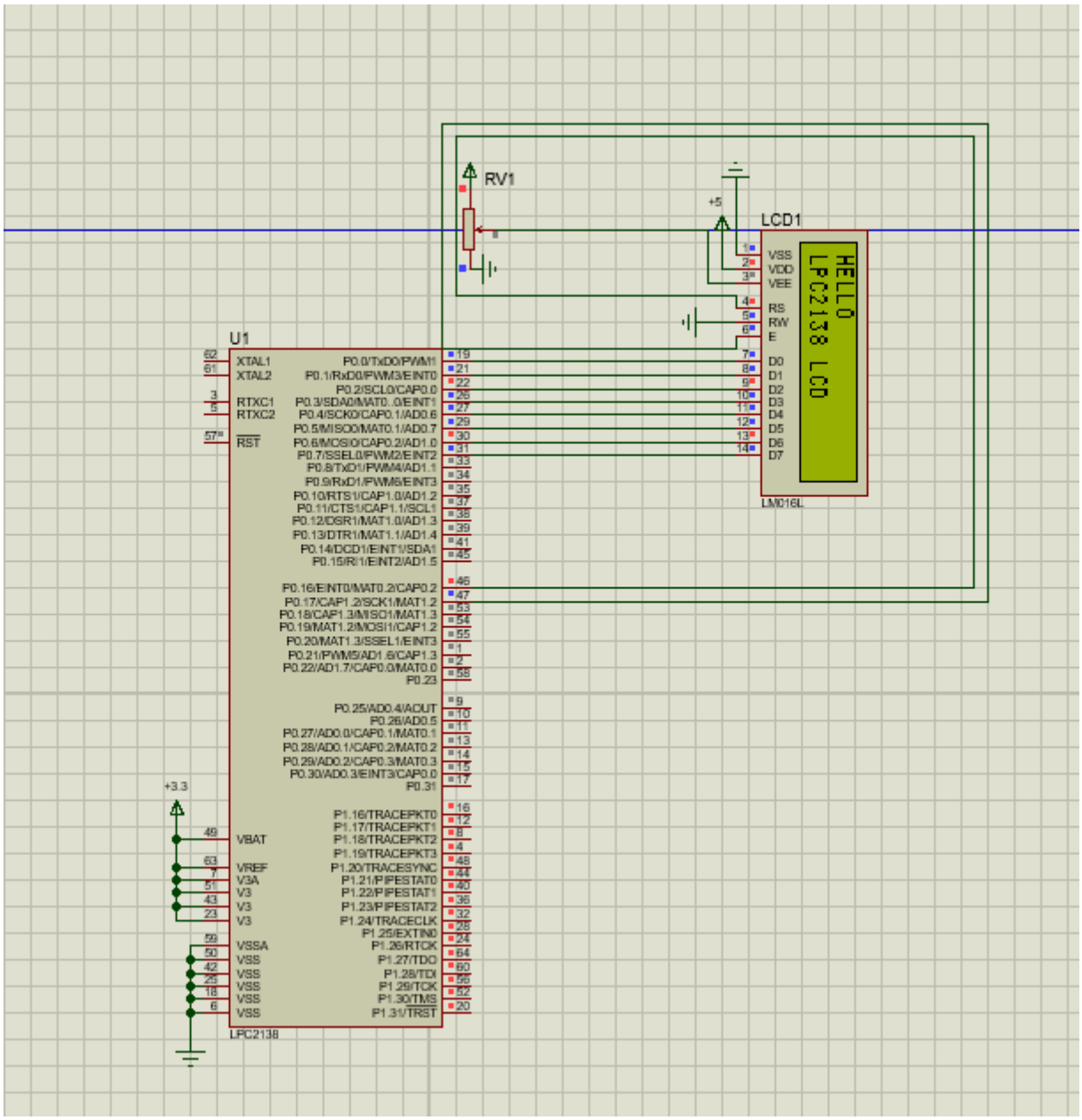
    lcd_string("LPC2138 LCD");

    while(1);
}
```

OUTPUT WINDOW:


Registers
Register
Value
Current
R0 0x40000060
R1 0x40000060
R2 0x40000060
R3 0x40000060
R4 0x00000000
R5 0x40000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000328
R11 0x00000000
R12 0x000002CC
R13 (SP) 0x40000460
R14 (LR) 0x000000C0
R15 (PC) 0x000002CC
CPSR 0x600000D3
SPSR 0x00000000
User/System
Fast Interrupt
Interrupt
Supervisor
Abort
Undefined
Internal
PC $ 0x000002CC
Mode Supervisor
States 866
Sec 0.00007217
Disassembly
62: IODIR0 = LCD_DATA | RS | EN;
63:
0x000002CC E59F0028 LDR R0,[PC,#0x0028]
0x000002D0 E59F1028 LDR R1,[PC,#0x0028]
0x000002D4 E5810008 STR R0,[R1,#0x0008]
64: lcd_init();
Project
Project: lab4
Target 1
Source Group 1
Startup.s
lab4.c
lab4.c
#include <LPC21xx.h>
#define LCD_DATA 0x000000FF // P0.0 to P0.7
#define RS (1<<16)
#define EN (1<<17)
void delay(unsigned int t)
{
unsigned int i,j;
for(i=0;i<t;i++)
for(j=0;j<600;j++);
}
void lcd_cmd(unsigned char cmd)
{
IOCLR0 = LCD_DATA;
IOSET0 = cmd;
IOCLR0 = RS; // Command mode
IOSET0 = EN;
delay(2);
IOCLR0 = EN;
delay(5);
}
void lcd_data(unsigned char data)
{
IOCLR0 = LCD_DATA;
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ch.sc.u4cse24109\\Desktop\\CH.SC.U4CSE24109\\Lab-4\\Objects\\lab4.axf"
*** Restricted Version with 32768 Byte Code Size Limit
*** Currently used: 808 Bytes (2%)
ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE DEFINE DIR Display Enter EVALuate
Call Stack + Locals
Name
Location/V...
Type
main 0x000002CC int f()
Call Stack + Locals
Memory 1
Show or hide the Disassembly Window
Real-Time Agent: Target Reset

OUTPUT:


RV1
+5
LCD1
VSS
VDD
VEE
RS
RW
E
D0
D1
D2
D3
D4
D5
D6
D7
HELLO
LPC2138 LCD
LM016L
U1
XTAL1
XTAL2
RTXC1
RTXC2
RST
VBAT
VREF
V3A
V3
V3
V3
VSSA
VSS
VSS
VSS
VSS
VSS
+3.3
LPC2138